

MINOR PROJECT REPORT

Project Title: Configuring and Securing a Network

Infrastructure Subtitle: Minor Project – Cyber Security

Student Name: Sourav Verma

Course / Branch: B.Tech (Cyber Security)

Acknowledgement

I would like to sincerely thank everyone who supported me during the completion of this minor project. I am grateful to my mentors and faculty members for their guidance, suggestions, and constant encouragement throughout the work. I also appreciate the facilities and lab environment provided, which helped me practically design, configure, and secure a network using Cisco Packet Tracer, Windows, and Kali Linux. Their support played an important role in helping me understand both the technical and practical aspects of network security.

Abstract

This project focuses on the practical design and security of a small enterprise-style network in a virtual environment. Using Cisco Packet Tracer, routers, switches, and end devices were configured to simulate a real-world network setup. Important security and performance features such as Access Control Lists (ACLs), Virtual LANs (VLANs), Quality of Service (QoS), and Secure Shell (SSH) were implemented to control access, improve traffic handling, and allow secure device management. The network was tested using connectivity checks, scanning tools, and live traffic monitoring to ensure that it works reliably and follows basic security practices.

1. Introduction

In today's digital world, a secure and reliable network is essential for communication, data sharing, and system management. Organizations depend on their networks to run daily operations, which makes them a common target for security threats. This minor project aims to build a small-scale enterprise network in a virtual lab and apply basic security and performance techniques to protect it. The project covers practical concepts such as IP addressing, VLAN-based segmentation, firewall rules using ACLs, traffic prioritization through QoS, and secure remote access using SSH. The main goal is to understand how these concepts work together in a real networking environment.

2. Assessment and Planning

2.1 Network Diagram

The proposed network consists of a router, a switch, and multiple end devices connected within a local area network. The router works as the main gateway and handles security and traffic control, while the switch is used to connect end devices and manage VLAN-based segmentation.

Network Topology:

PC1 and PC2 are connected to a switch. The switch is connected to a router, which manages routing, security rules, and quality of service policies for the entire network.

2.2 Requirements Gathering

Security Requirements:

- The network should block unauthorized access using firewall rules.
- Administrative access to network devices should be secure.
- User access should be controlled and monitored.

Performance Requirements:

- Important traffic should be given priority over normal traffic.
- Communication between devices should be smooth and reliable.

Scalability Requirements:

- The design should allow more users and VLANs to be added in the future without major changes.

3. Hardware and Software Selection

3.1 Hardware Components

Cisco Router (2911): Routing, firewall configuration, QoS, and secure management

Cisco Switch (2960): VLAN creation and device connectivity

PC1, PC2: End-user systems

3.2 Software Components

Cisco Packet Tracer for network simulation and configuration

Windows operating system for host-based testing

Kali Linux virtual machine for security testing and monitoring

Command Line Interface (CLI) for configuration and verification

4. Network Configuration

4.1 IP Addressing Scheme

The network uses a private IPv4 addressing scheme for internal communication. Each device is assigned a unique IP address within the same subnet to allow proper connectivity.

Router (Gi0/0): 192.168.10.1 / 255.255.255.0

PC1: 192.168.10.10 / 255.255.255.0 / Gateway: 192.168.10.1

PC2: 192.168.10.11 / 255.255.255.0 / Gateway: 192.168.10.1

4.2 VLAN Configuration

To logically separate users and improve network organization, two VLANs were created:

VLAN 10 – STAFF

VLAN 20 – ADMIN

Switch ports were assigned to the respective VLANs so that devices in different groups remain logically separated even though they are connected to the same physical switch.

4.3 Routing and Switching

The router is configured as the default gateway for all devices and is responsible for forwarding traffic and applying security and performance policies. The switch operates in access mode for end devices and ensures proper communication within and across VLANs.

4.4 Quality of Service (QoS)

A QoS policy was applied on the router to give higher priority to important traffic such as control and testing packets. This helps ensure better performance and lower delay for critical communication, especially when the network is under load.

5. Network Security Implementation

5.1 Firewall Configuration (ACL)

An Access Control List (ACL) was configured on the router to control which traffic is allowed to enter and leave the network. The rules were designed to allow traffic from the internal network while blocking all unnecessary or unauthorized traffic.

Example Rules:

- Allow traffic from the internal network (192.168.10.0/24)
- Deny all other traffic by default

5.2 Access Control and Secure Management

Secure Shell (SSH) was enabled on the router to allow encrypted remote access for administrators. A local user account was created so that only authorized users can log in and manage the network device.

5.3 Encryption

SSH version 2 was used to ensure that all management traffic between the administrator and the router is encrypted, protecting login credentials and configuration data from being exposed.

6. Monitoring and Maintenance

6.1 Network Monitoring

To verify that the network is running properly, monitoring commands such as `show ip interface brief`, `show access-lists`, and `show policy-map interface` were used. These commands help check interface status, firewall rules, and QoS policies.

6.2 Patch Management

Regular updates and backups are recommended to keep the network secure and stable. This includes saving configuration files and applying firmware updates whenever required.

6.3 Incident Response

If a security issue is detected, firewall rules can be modified to block suspicious traffic. Monitoring outputs and logs can then be reviewed to identify the source of the problem and take corrective action.

7. Testing and Validation

7.1 Performance Testing (ICMP Ping)

Ping tests were performed from the Kali Linux virtual machine to the target host at 192.168.56.1. The continuous echo replies and low response time confirmed that the network connection is stable and that devices can communicate without major delays.

7.2 Network Discovery and Port Scanning (Nmap)

Nmap was used to scan the 192.168.56.0/24 network to discover active hosts and identify open services. The scan showed that only limited ports were accessible, which indicates that firewall

rules are working as expected.

7.3 Windows Host Firewall Validation

On the Windows host system, firewall rules were added to allow ICMP (Ping) and HTTP traffic. The successful execution of these rules confirmed that the host-based firewall is properly configured and under administrative control.

7.4 Network Interface Verification (Kali Linux)

The `ip a` command was used to verify that the Kali Linux virtual machine had both a NAT interface and a host-only interface. This confirmed that the system was correctly connected to the testing network as well as the internal lab environment.

7.5 Traffic Monitoring (Wireshark)

Wireshark was used to capture ICMP traffic on the host-only interface. The captured packets showed echo requests and replies, along with details from the Ethernet, IP, and ICMP layers. This confirmed that live traffic could be monitored and analyzed for troubleshooting and security purposes.

8. Documentation and Training

All configurations, IP schemes, VLAN details, firewall rules, and security settings were documented for future reference. Basic training was provided on secure password usage, safe network practices, and simple troubleshooting steps to ensure that users and administrators can handle common network issues.

9. Compliance and Audit

The network follows basic security principles aligned with common standards such as ISO 27001 and general data protection practices. Regular reviews of firewall rules, user access, and system configurations help ensure that security policies remain effective over time.

10. Results and Discussion

The network performed as expected during testing. VLAN segmentation improved organization and reduced unnecessary traffic between different user groups. The ACL-based firewall successfully limited access to only required services, and the QoS policy ensured that important traffic was given priority. Secure access through SSH added an extra layer of protection for administrative tasks. Overall, the results show that the network is stable, secure, and flexible enough to support future improvements.

11. Conclusion and Future Scope

Conclusion

This minor project successfully demonstrates how a basic enterprise-style network can be designed and secured in a virtual environment. By using VLANs, ACLs, QoS, and SSH, the project shows how performance and security can be managed together. The practical testing and monitoring steps helped in understanding how real networks are configured and maintained.

Future Scope

- Integration of intrusion detection or prevention systems
- Implementation of VPN access for remote users
- Use of centralized monitoring and logging tools
- Expansion of the network to include multiple routers and locations

12. References

Cisco Networking Academy – Packet Tracer Documentation

ISO/IEC 27001 Information Security Standards

Andrew S. Tanenbaum, Computer Networks

Appendix – Commands Used

Router Commands:

enable

configure terminal

interface gigabitEthernet0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

access-list 100 permit ip 192.168.10.0 0.0.0.255 any

access-list 100 deny ip any any

access-list 101 permit icmp any any

class-map match-all IMPORTANT

match access-group 101

policy-map QOS-POLICY

class IMPORTANT

priority 1000

hostname SecureRouter

ip domain-name lab.local

crypto key generate rsa

username admin secret Admin@123

line vty 0 4

login local

transport input ssh

Switch Commands:

enable

configure terminal

vlan 10

name STAFF

vlan 20

name ADMIN

interface fastEthernet0/1

switchport mode access

switchport access vlan 10

interface fastEthernet0/2

switchport mode access

switchport access vlan 20

End of Report